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Roll No .....

**EC-7012 (GS)**  
**B.E. VII Semester** Examination, June 2020  
**Grading System (GS)**  
**Digital Image Processing**  
*Time : Three Hours*

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. Differentiate between gray level slicing and bit plane slicing.
2. Show that the  $N \times N$  sine transform is orthogonal. Verify your proof for the case  $N = 3$ .
3. The  $3 \times 3$  mask shown below is frequently used to compute the derivative in the x direction at each point in an image. Give an ALU procedure to implement this operation. Given an image is -

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

4. a) Write short notes:  
i) Walsh transform                      ii) Hadamard transform  
b) Explain K.L transform in detail.
5. What are the fundamental steps in digital image processing system? Explain each steps in detail.
6. a) Explain run length encoding in detail.  
b) What is image compression? Explain Huffman coding type variable length coding compression schemes.

OR

Show that the continuous Gaussian lowpass filter in frequency domain has the transfer function  $H(u, v), Ae^{-(u^2+v^2)/2}$ .

7. Find 2D Hadamard transform of given image matrix —

$$f(m,n) = \begin{vmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 1 & 2 & 1 \end{vmatrix}$$

8. Write short note:(Any two)  
i) Huffman coding                      ii) Digital image watermarking  
iii) Inverse filtering                      iv) Noise modes

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